Pravallika Nalluri
Independent Researcher, USA
Email: pravallikanalluri9848@gmail.com

October 24, 2026

The Editor-in-Chief
Knowledge-Based Systems
Elsevier

Dear Editor-in-Chief,

Please consider the enclosed manuscript for publication as a Short Communication in *Knowledge-Based Systems*.

Title: Knowledge-Based Cost-Aware Decision Support for Credit Classification: Separating Model Selection from Decision Thresholds.

A predicted probability is not a credit decision. This short communication separates which model to keep from which threshold turns that probability into an action. The knowledge component is compact and inspectable: one false-negative cost ratio is fixed before any test account is scored, and the operational object is the pair (r, t). Prediction-first selection fixes the model by validation F1 at threshold 0.50 and then tunes only the threshold. Decision-first selection chooses the model and the threshold together by validation cost. The test split is scored only after both rules are locked. On the public UCI Default of Credit Card Clients file, leaving threshold 0.50 accounts for more of the cost improvement than the choice between these two selection rules.

The manuscript is submitted to *Knowledge-Based Systems* because the contribution is a knowledge-based operating rule for a credit decision. The preference r is explicit organizational knowledge. The policy t is the threshold that turns a learned probability into an action.

I am the sole author. The manuscript is original, has not been published, and is not under consideration by another journal. I approve this submission.

Declaration of competing interest. The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used Cursor (Grok) to assist with code structure, experiment implementation, and language editing of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability. The Default of Credit Card Clients file is the UCI Machine Learning Repository dataset introduced by Yeh and Lien (https://doi.org/10.24432/C55S3H). The script that regenerates the tables and figures is `run.py` at https://github.com/pravallika-nalluri-dba/kbs-cost-aware-classification.

The manuscript source file submitted with this letter is `Nalluri_KBS_SourceFile.docx`. Thank you for your consideration.

Sincerely,

Pravallika Nalluri
Corresponding author
pravallikanalluri9848@gmail.com